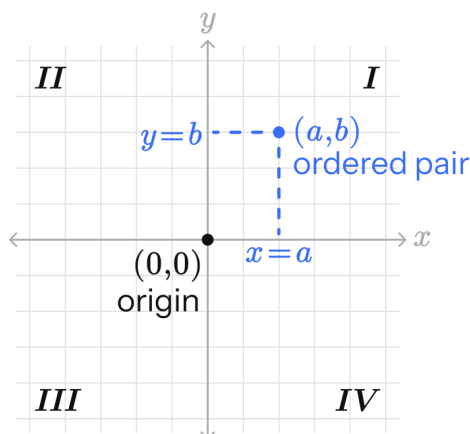


1.1 Intro to Graphing



- **x-intercept** $(\#,0)$ — where line hits x-axis; $y=0$
- **y-intercept** $(0,\#)$ — where line hits y-axis; $x=0$
- **slope** m — the steepness and direction of the line
 $\text{rise/run} = \text{change in } y / \text{change in } x = \text{height/width}$

Things to Know

- negating x FLIPS the line across the y -axis
- **SLOPE TYPES**: positive (uphill $+/+$ or $-/-$); negative (downhill $+/-$ or $-/+$); zero (horizontal $0/\#$); infinite (vertical $\#/0$)

Techniques/Skills

- **finding x-intercept**: plug in $y=0$ and solve for x
- **finding y-intercept**: plug in $x=0$ and solve for y

1.2 Linear

- **expression** — consists of numbers, variables, and arithmetic operations
- **equation** — two expressions set equal
- **degree** — highest exponent on the variable
- **linear** — expression/equation of degree 1

Things to Know

- parallel lines: SAME slope
- perpendicular: NEGATIVE & RECIPROCAL slope

Techniques/Skills

- **graphing linear equations**: find intercepts, connect the dots
- **finding equation from points**: calculate slope using points; plug a point into and slope into $y=mx+b$ to solve for b

1.3 Linear Systems

- **system of equations** — a collection of equations that must be true simultaneously

Things to Know

- linear sys can have EXACTLY 1 solution, INFINITE solutions (same line), or NO solutions (parallel)

Techniques/Skills

- **check solution**: plug into BOTH equations
- **solving by substitution**: isolate a variable, use the result to replace it in the OTHER equation; solve variable; back-substitute to solve for the other
- **solving by elimination**: pick variable to eliminate and scale; add equations to cancel; solve variable; back-substitute to solve for the other

1.4 Word Problems

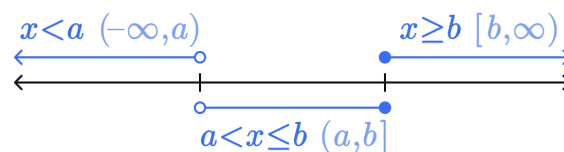
Things to Know

- A is P% MORE than B: $A = (1 + P/100) \times B$
 A is P% LESS than B: $A = (1 - P/100) \times B$
 $\rightarrow$ P will be different if you swap A and B!!!!!!

Techniques/Skills

- **translating words to math**: identify unknowns & assign variables; make equations (*totals!*); solve
- **catching wrong answers**: do the numbers sound correct? are they way too high/low? do they match the problem's given information?

1.5 Linear Inequalities



- **linear inequality** — two expressions combined with an inequality sign ("compound" = three)

Things to Know

- inequality vs. graph vs. interval notation (all with open vs. closed bounds)
- FLIP sign when you $\times$ or $\div$ by negative number

Techniques/Skills

- **solving inequalities**: isolate variable on one side (compound = isolate variable in middle)